# A Text-Based Geopolitical Alignment Index from Foreign Ministry Press Releases

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# Introduction

Geopolitical alignment plays a central role in international trade, capital flows, and technology diffusion. Yet the empirical measures of geopolitical alignment remain limited.

The dominant approach relies on a geopolitical measure created using the similarity of the voting patterns at the UN General Assembly (Bailey, Strezhnev & Voeten, 2017). While valuable, UN voting measures are centered around topics presented at the UNGA and they do not capture any shifts in a country's narrative. This creates a need for an updated measure of geopolitical alignment which captures those topics. So, we propose a new measure using text analysis.

Recent research has turned to text-based methods to quantify geopolitical phenomena. For example, Caldara and Iacoviello (2022) construct a Geopolitical Risk (GPR) index using newspaper-based word counts. Additionally, the preliminary draft of Iacoviello and Tang (2026) uses an AI-enhanced geopolitical risk index that improves on the earlier 2022 paper. The authors claim that this creates a more accurate, context-aware, and high-frequency measure of geopolitical risk from newspaper text.

However, these indices measure global geopolitical risk rather than bilateral alignment specifically. This paper thus addresses the gap by proposing a new text-based measure of bilateral geopolitical alignment derived directly from official diplomatic communications. This paper builds on the existing literature by developing two approaches to measuring geopolitical alignment from official diplomatic statements:

- A contextual LLM-based evaluator
- A rule-based usage-weighted token methodology

The contribution lies not only in constructing a new index, but in directly comparing AI-based contextual scoring with interpretable rule-based text metrics.

# Research Question

This paper addresses two central questions:

1. Can official diplomatic documents provide a reliable measure of geopolitical alignment?
2. How does an LLM-based contextual alignment measure compare to a rule-based usage-weighted count-based measure?

# 3. Data

### 3.1 Source

This paper uses official press releases published by national foreign ministries as its primary data source. Until now, we have collected official press releases from the foreign or external affairs ministries of countries like India, China, Canada, Norway, the United Kingdom, Singapore, Italy and Spain.

Each press release includes structured data such as the publication date and title, along with the full text of the statement. The documents typically reference foreign governments, political leaders, bilateral meetings, strategic dialogues, joint statements, and other diplomatic engagements. These releases therefore provide direct insight into how governments publicly convey their respective relationships with other states.

Due to coding constraints, we had to decide on the coverage period depending on website accessibility. For India, we were able to download documents going back to 2000. As it is the initial stages of the project, we decided to download for the G20 countries. This serves to validate the data extraction pipeline and alignment construction methodology.

### 3.2 Unit of Analysis

The primary unit of analysis is: Origin–Target Country–Time Period

For example:

- India–United States in a specific date
- India–China in a specific date

Country mentions are standardized to ISO-3 codes using automated country mapping.

## 4. Methodology

The paper constructs two alignment measures.

### Method 1: Usage-Weighted Geopolitical Alignment Index

This approach builds on the dictionary-based text measurement tradition (Gentzkow, Shapiro & Taddy, 2019).

The approach constructs a rule-based measure of bilateral geopolitical alignment using proximity-aware sentiment attribution and usage weighting. The goal is to quantify how

positively or negatively an origin country publicly characterizes a target country in its official diplomatic statements.

A naive word-count approach would assign all positive and negative words in a document to every country mentioned. However, diplomatic statements frequently reference multiple countries within the same sentence, creating misattribution risk. To address this, sentiment words are assigned to a target country only if they are closer within the sentence to that country than to any other country.

Sentiment is identified using positive diplomatic phrases (such as “strategic,” “allied,” “cooperative”) from negative terms (such as “condemnation,” “violation,” “aggression”); more are in the appendix. To reduce the influence of generic language, each word is weighted by its inverse document frequency (IDF) across country-pairs, giving greater weight to terms that are rare and thus more informative.

### Method 2: LLM-Based Geopolitical Alignment Index

This method is inspired from Iacoviello and Tang (2026) which uses ChatGPT 4o mini to create a more accurate, context-aware, and high-frequency measure of geopolitical risk from newspaper text.

### Implementation

Foreign Ministry Press releases are uploaded for each country and the prompt mentioned in the appendix is used. In each document mentioning a target country that the LLM receives, It identifies the following objects:

- The text
- Origin country
- Target country

The model outputs an alignment score between -1 and +1. Compared to dictionary-based geopolitical risk measures (Caldara & Iacoviello, 2018), the LLM approach evaluates the release based on the context, avoids rigid word lists unlike method 1, and detects subtle changes in the diplomatic tone.

## 5. Result

We apply both alignment measures to official diplomatic text to evaluate how well they capture bilateral positioning. The first set of results uses the **usage-weighted proximity-based index**

(Method #1). The second uses the **LLM-based alignment score** (Method #2), as generated by ChatGPT 5.2 for a fixed date.

## 5.1 Usage-Weighted Alignment Scores (Canada, 2024)

According to the usage weighted method, here are the top 5 closely aligned countries with Canada:

| origin | country | iso3c | n_docs | pos_total | neg_total | net_total |
|---|---|---|---|---|---|---|
| **CAN** | United States | USA | 17 | 73.44 | 25.76 | 42.55 |
| **CAN** | Ukraine | UKR | 32 | 392.03 | 298.03 | 34.39 |
| **CAN** | Venezuela | VEN | 5 | 26.27 | 0 | 26.27 |
| **CAN** | Hong Kong SAR China | HKG | 4 | 22.28 | 0 | 22.28 |
| **CAN** | Mexico | MEX | 6 | 19.07 | 0 | 19.07 |

These net totals show Canada's rhetoric towards the United States and Ukraine has been positive. This could be due to their historical ties with the US and recent positive rhetoric in support of Ukraine amidst the Russian invasion. Smaller diplomatic outputs with Mexico, Venezuela and Hong Kong also register positive net alignment.

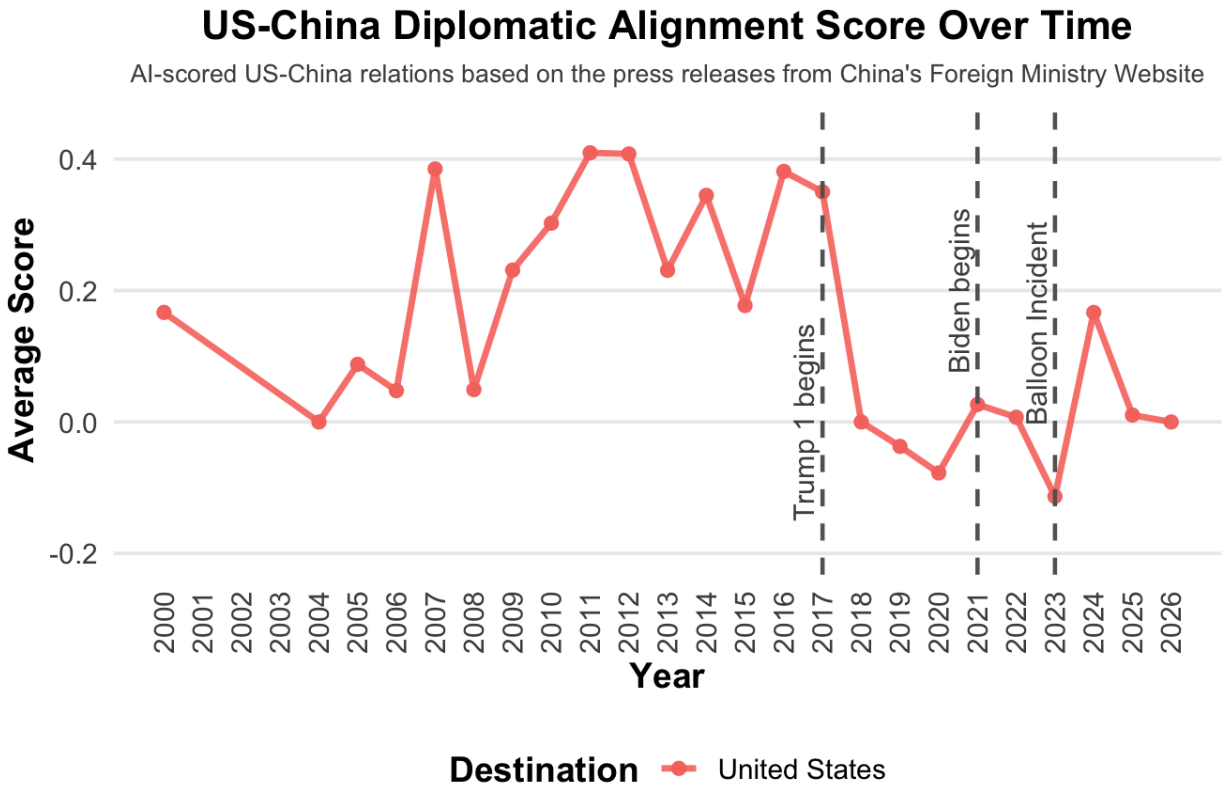
Top 5 least-closely aligned/adversarial countries with Canada:

| origin | country | iso3c | n_docs | pos_total | neg_total | net_total |
|---|---|---|---|---|---|---|
| **CAN** | China | CHN | 2 | 0 | 18.15 | -21.78 |
| **CAN** | Iran | IRN | 5 | 2.89 | 25.82 | -28.09 |
| **CAN** | Syria | SYR | 4 | 4.77 | 38.51 | -41.44 |
| **CAN** | Israel | ISR | 11 | 13.52 | 83.67 | -86.88 |
| **CAN** | Russia | RUS | 28 | 93.51 | 510.52 | -519.17 |

Negative net totals look at a greater use of negative words in statements when talking about the countries above. For example, Canada has publicly condemned Iranian human rights abuses, Chinese interference in Canada, and Russia’s actions in Ukraine, including multilateral forums and sanctions announcements. In the above tables, n_docs mean the number of documents mentioning a country, pos_total gives the total positive words score, neg_total gives the total negative score, and net_total is the sum of positive and negative multiplied by an additional penalty for the negative words

### 5.2 LLM-Based Alignment Scores (China and United States, 2000-2025)

The following time series alignment scores were generated by GPT-5-Nano through China's Ministry of Foreign Affairs's official publication and statement relating to the United States of America from **2000 to 2025**. This graph reflects the evaluation of the bilateral relationship between China and the US. This shows certain sharp jumps in US-China relations. For example, in 2023, the US-China relations soured due the "balloon incident" where the US shot down a Chinese balloon flying over its territory. This incident is captured in the trends.



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# Appendix:

**For method #1**

```
pos_unigrams <-
tolower(stringi::stri_trans_general(c("strategic","ally","allied","friendship","partnership","partne
rs","cooperation","collaboration","commitment","committed","shared","values","bilateral","defe
nce","security","prosperity","stability","support","enduring","longstanding","deepening","strengt
hen","strengthening","enhance","enhancing","upgrade","upgrading","mutual","framework","roa
dmap","constructive","positive","productive","trusted","dialogue","agreement","agreements","tr
eaty","partnerships","solidarity","coordination","joint","coordinated","alignment","convergence"
,"affinity","friend"), "Latin-ASCII"))

pos_bigrams <- tolower(stringi::stri_trans_general(c("strategic partnership","comprehensive
strategic","time tested","time tested relationship","close partnership","defence
cooperation","security cooperation","economic partnership","strategic dialogue","shared
values","like minded","rules based","free trade","joint statement","joint declaration","mutual
prosperity","mutual security","long standing","enhanced partnership","long term
partnership","enduring partnership”), "Latin-ASCII"))

pos_trigrams <- tolower(stringi::stri_trans_general(c("comprehensive strategic
partnership","rules based international","time tested relationship","people to people","free trade
agreement","long term strategic”), "Latin-ASCII"))

neg_unigrams <-
tolower(stringi::stri_trans_general(c("sanction","sanctions","condemn","condemns","condemned
","condemning","illegal","unlawful","aggression","aggressive","attack","attacks","invade","inva
sion","occupy","occupation","annexation","violation","violations","breach","breaches","detain",
"detained","detention","expel","expulsion","ban","banned","proscribe","proscription","terrorist",
"hostile","poison","assassination","spy","espionage","cyberattack","cyberattacks","interference",
"coercion","threat","threats","escalation","escalatory","destabilising","malign","ransomware","di
sinformation","propaganda","atrocity","war","warfare","warcrime","warcrimes"),"Latin-ASCII")
)

neg_bigrams <- tolower(stringi::stri_trans_general(c("impose sanctions","sanctions on","illegal
invasion","unprovoked aggression","war crimes","grave violations","gross violations","human
rights violations","violation of","breach of","threat to","threats to","state sponsored","malign
activity","cyber attack","cyber attacks","chemical weapons","use of","accountable for","hold
accountable"), "Latin-ASCII"))

neg_trigrams <- tolower(stringi::stri_trans_general(c("in violation of","gross violations
of","breach of international","use of chemical","responsible for poisoning","attempted
assassination with"), "Latin-ASCII"))
```

**For method #2**

The prompt we used to calculate geopolitical alignment using ChatGPT:

“You are evaluating diplomatic language in an official Chinese foreign ministry document.

Does this text indicate cooperative alignment, neutral positioning, or adversarial positioning between {origin} and {target}?

IMPORTANT:

- Focus only on language relevant to the relationship between {origin} and {target}.

- This language is often found in meeting summaries, joint statements, press conferences, spokesperson remarks, or bilateral cooperation sections.

- If the document discusses {target} separately from other countries, evaluate ONLY the language about {target}.

- If {target} is grouped together with other countries or regions in a single statement and not separately identified, evaluate the grouped language.

- If the text refers only to {target}, treat it as a bilateral statement.

Assign a score from:

-1.0 = strongly adversarial

0 = neutral

+1.0 = strongly cooperative

Consider:

- tone

- commitments

- agreements

- strategic partnerships

- diplomatic support

- disputes

- military tensions

- sanctions

- accusations

- condemnations

- opposition to another state's actions

- references to sovereignty, territorial issues, or interference

Scoring guidance:

- Use positive scores for clear cooperation, partnership, support, or deepening ties.

- Use negative scores for condemnation, conflict, protest, or hostile positioning.

- Use 0 for procedural, factual, or ambiguous language with no clear directional alignment.

- If both positive and negative language appear, score the net diplomatic posture toward {target}.

- Do not over-weight generic diplomatic courtesy unless it clearly signals substantive cooperation.

Output only a numeric score.

Temperature = 0.

TEXT:

{chunk}”